

Question:

A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?

Reasoning + Answer:

<think> Set up two variables: b for blue bolts, w for white bolts. The robe takes 2 b bolts of blue fiber. The robe takes half the number of bolts of white fiber of blue fiber, or $w = b / 2$ bolts of white fiber. The total number of bolts is $2b + w$ or $2b + b / 2$. Combine like terms: $2b + b / 2 = 5b / 2$. Since the total number of bolts is a whole number, $b / 2$ has to be a whole number. So, b has to be even, and in this case we need to try 2 and 4. If $b = 2$: 2 blue bolts * 2 = 4 blue bolts; 2 white bolts * 1 = 2 white bolts; 4 + 2 = 6 bolts. If $b = 4$: 2 blue bolts * 2 = 4 blue bolts; 2 white bolts * 2 = 4 white bolts; 4 + 4 = 8 bolts. Only 8 is even. <answer> 8

Reward score